

Jeremy Hsu

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EDUCATION

Pennsylvania State University

Bachelor of Science, Computer Science (College of Engineering)

University Park, PA

Aug. 2025 – Dec. 2028

- **Relevant Coursework:** Data Structures, Discrete Mathematics for Computer Science, Linear Algebra, Multivariable Calculus & Vector Analysis, Applied Regression Analysis

RESEARCH EXPERIENCE

Undergraduate Researcher

Locomotion in Biology and Robotics (LiBR) Lab, Penn State

Aug. 2026 – Present

University Park, PA

- Researching locomotion in unconventional body plans — multi-legged, elongated, and cable-driven robots — under Dr. Baxi Chong
- Modeling scaling laws for computational versus mechanical cost of transport

Research Intern

Robot Perception and Navigation Group, University of Delaware

June 2024 – July 2024

Newark, DE

- Built a 2D SLAM pipeline on a TurtleBot3 using ROS 2 and Google Cartographer in C++ to map an indoor lab environment
- Integrated Nav2 for autonomous waypoint navigation on the resulting map
- Presented navigation and sensing results to graduate researchers and faculty

ENGINEERING EXPERIENCE

Founder & Technical Lead

FTC #16468 Green Lemons Robotics

June 2019 – Apr. 2025

West Chester, PA

- Achieved #1 OPR globally among 8,000+ teams at the 2024 FTC World Championships
- Founded and led a competitive robotics team across six seasons, owning recruiting, training, and technical direction for 14 members
- Developed autonomous control software in Java, including vision-based navigation, odometry, and path-planning algorithms
- Designed and fabricated modular drivetrain and manipulator mechanisms using CAD and custom metalwork
- Mentored junior members in programming, electronics, and mechanical design

PROJECTS

Kintoun | Python

Aug. 2025 – Present

- Built an event-driven backtester for a 15-minute opening-range breakout on NQ futures 1-minute bars
- Engineered the data layer for correctness: DST-aware US/Eastern alignment and exclusion of identified feed-degradation days
- Vectorized indicators in NumPy/pandas while keeping execution in an event-driven loop
- Simulated 2022-2025 across 1,062 trades: profit factor 1.47 at a 33.9% win rate, Sharpe 2.28, max drawdown -8.5%

ADDITIONAL EXPERIENCE

Founder & Quantitative Researcher

Weaver Capital LP

Jan. 2026 – Present

West Chester, PA

- Engineered algorithmic execution systems in Python and C#, featuring automated strategy synchronization, risk controls, and sub-second order handling
- Built portfolio optimization pipelines to allocate capital dynamically across uncorrelated strategy sleeves

TECHNICAL SKILLS

Languages: Python, C++, Java, C#

Robotics: ROS 2, Cartographer SLAM, Nav2, RViz2, LiDAR, OpenCV, path planning, control systems

ML / Statistics: PyTorch, TensorFlow, scikit-learn, Statsmodels

Data & Tools: NumPy, SciPy, pandas, Matplotlib, Git, Linux, Jupyter